

PROBLEM STATEMENT ID – SIH26075

Problem Statement Title -CAPACITY CONNECT

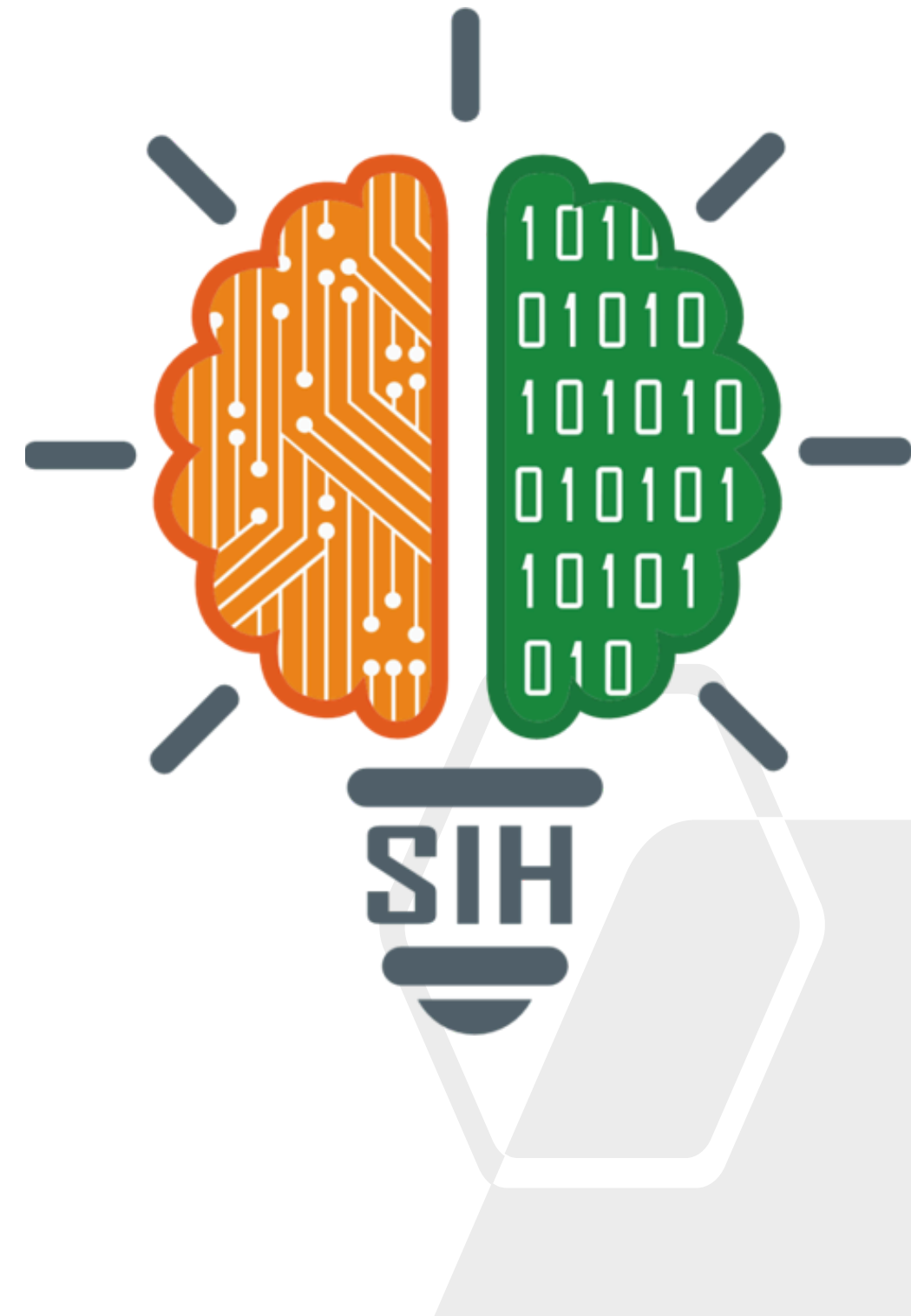
(A Digital Capacity Building and Learning Management Portal)

Theme- Smart Education

PS Category- Software

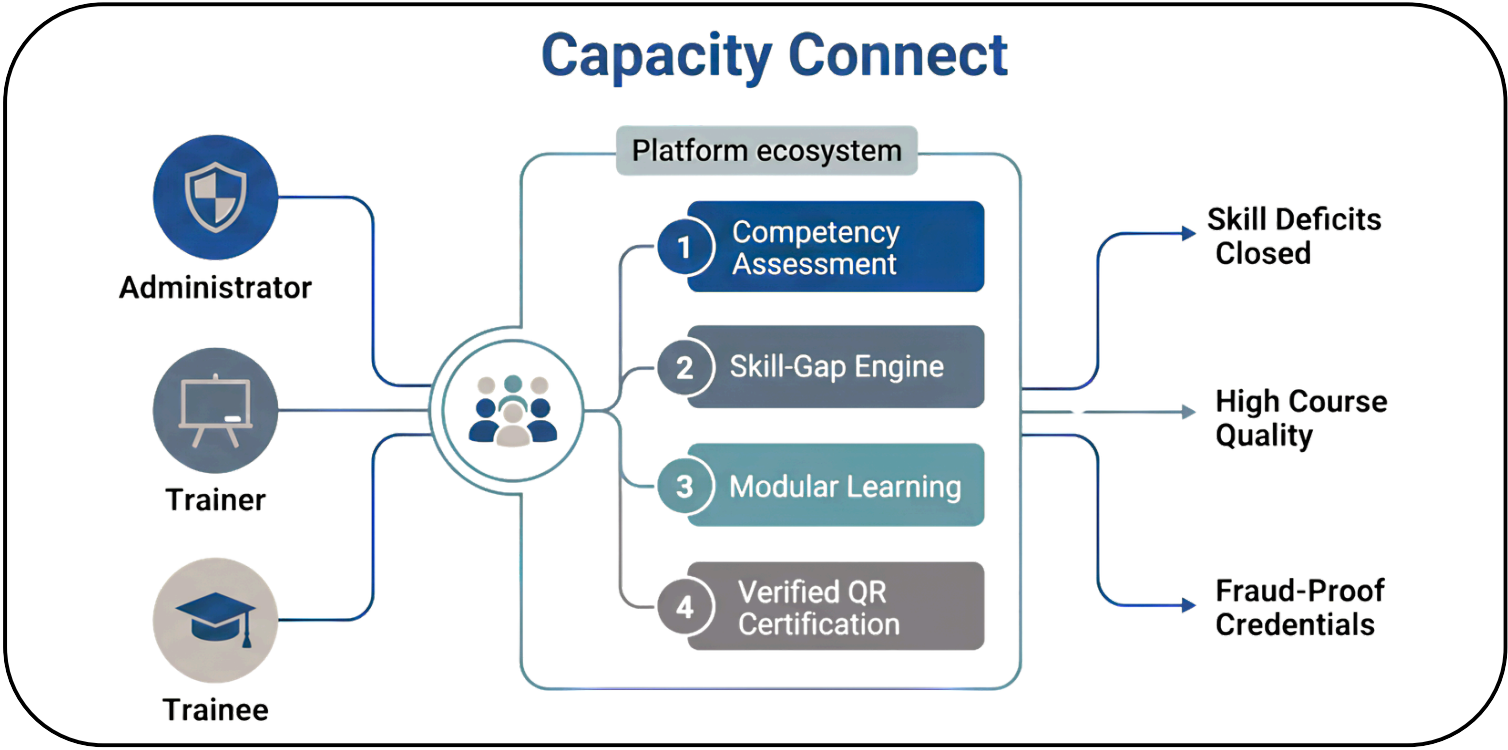
Team ID- 151096

Team Name- TEAM PRERAK



Detailed explanation of the proposed solution

- End-to-end digital capacity-building platform for Administrators, Trainers, and Trainees.
- Role-based access with a unified learning ecosystem.
- Automated competency benchmarking and real-time skill gap analysis.
- Interactive, structured learning and training modules.
- Live dashboards for progress tracking and performance analytics.
- Milestone assessments with verified credential management.



Existing Problems

Manual Spreadsheets & Paper Logs
Fragmented departmental data and high admin overhead

Untracked Skill Gaps
Generic training without diagnostic gap evaluation

Unverified Trainers & Fraud
Risk of unqualified educators and fake paper certificates

How it Addresses the Problem

Centralized Cloud Platform
Unified real-time data access with 80% administrative time saved

Automated Skill-Gap Engine
Algorithmic deficit computation and personalized learning roadmaps

Admin Vetting & QR Security
Mandatory syllabus approval and tamper-proof cryptographic certificates

Innovation and Uniqueness of the Solution (USP)

Automated Skill-Gap Engine

Algorithmic calculation of required vs current skill delta, categorizing deficits into High, Medium, and Low urgency tiers

6-Signal Weighted Matching

Multi-factor pairing formula scoring skill overlap, trainer ratings, schedule availability, and experience

Tamper-Proof QR Certification

Cryptographic SHA-256 validation hash embedded in certificates for instant scan-to-verify fraud prevention

Tri-Role Access Control (RBAC)

Enterprise-grade security ensuring strict isolation between Admin oversight, Trainer authoring, and Trainee learning

Technologies to be Used

- Frontend & UI:



Fast Server-Side Rendered (SSR) dynamic UI.



Type-safe, responsive component design.

- Backend & APIs:



Modular RESTful API architecture with DTO validation.

- Database & ORM:



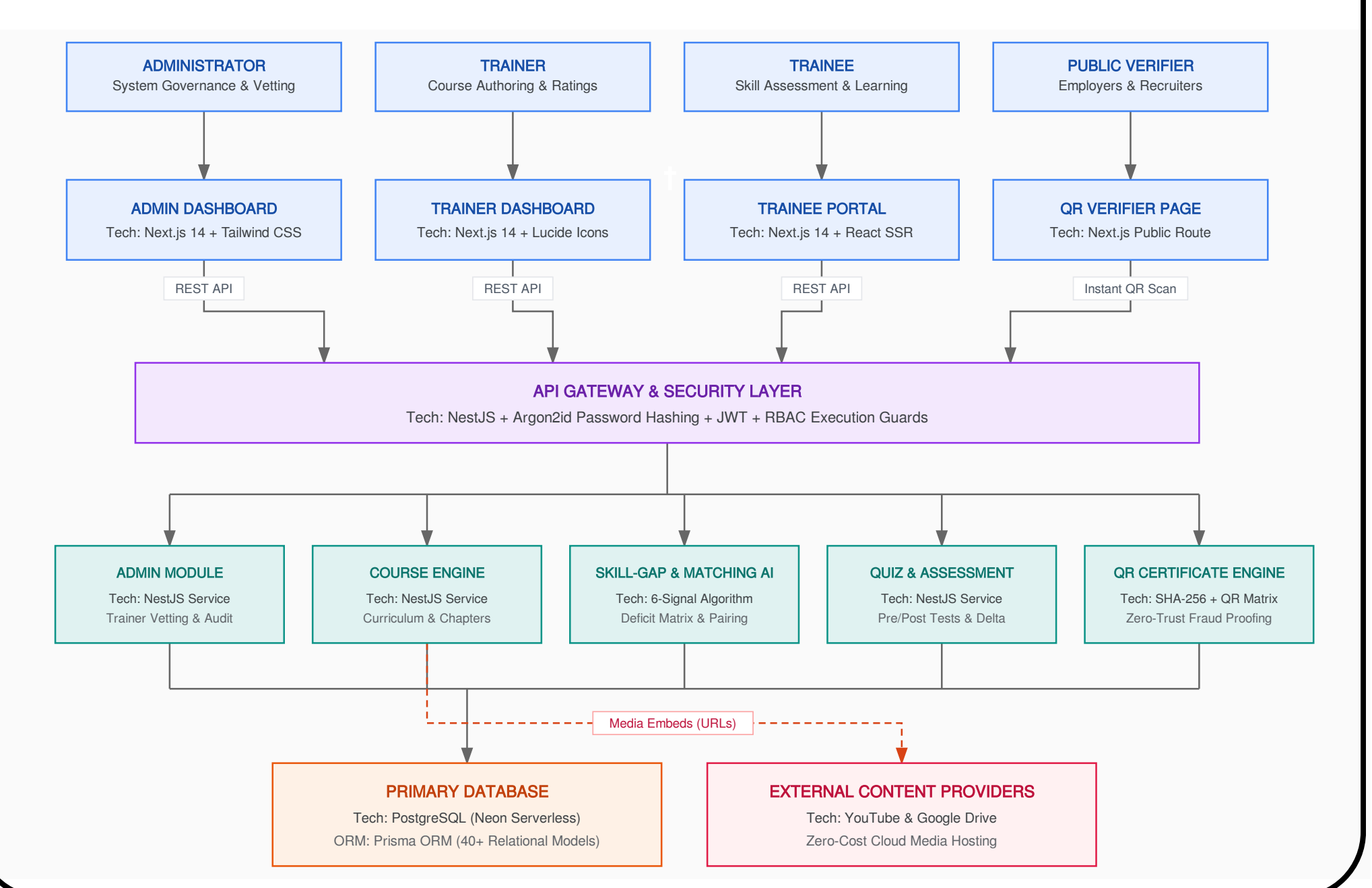
Relational database with type-safe queries & migrations.

- Security & Access Control:

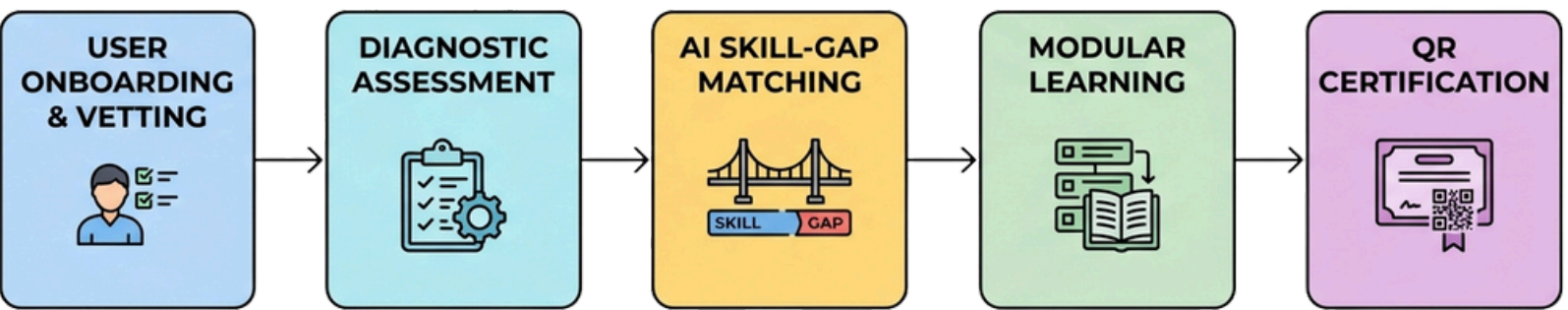


Cryptographic password hashing, token auth, and strict role isolation.

System Architecture & Data Flow Diagram



Methodology and process for implementation





TECHNICAL FEASIBILITY

Modern, secure and scalable architecture



Web-based platform
(No installation required)



Modular & scalable design
(Easy to extend / evolve)



Secure role-based access
(RBAC + JWT authentication)



Responsive across devices
(Mobile, Tablet, Desktop)



Reliable database architecture
(PostgreSQL + Prisma)



OPERATIONAL FEASIBILITY

Easy to deploy and manage



Zero-install web platform
(Accessible via browser)



Works across devices
(Mobile, Tablet, Desktop)



Centralized administration
(User & role management)



Trainer content management
(Library, lectures, materials)



User-friendly interface
(Simple & intuitive)



POTENTIAL CHALLENGES & RISKS

Identify, Prepare, Mitigate



Dynamic skill taxonomies
(Different institutions)



Concurrency spikes
(High traffic during exams/deadlines)



Trainer fraud & poor content
(Unverified credentials)



Privilege escalation
(Access to admin/evaluations)



STRATEGIES TO OVERCOME

Proactive measures for success



Standardized competency
(Framework & mapping)



Caching & DB optimization
(Handle traffic spikes)



Admin verification & ratings
(Ensure quality & authenticity)



RBAC + secure authentication
(Prevent unauthorized access)



KEY INNOVATIONS

What makes Capacity Connect unique?

01

Competency-Driven Learning

Training starts from actual skill gaps, not generic curriculum.

02

Role-Based Ecosystem

Admin + Trainer + Trainee in one unified platform.

03

Intelligent Trainer Matching

Recommend suitable trainers based on competency & subject requirements.

04

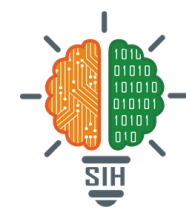
Rural-Ready Design

Lightweight architecture for low-bandwidth areas and wider accessibility.



TEAM PRERAK

IMPACT AND BENEFITS



SMART INDIA
HACKATHON
2026

1 Potential Impact on the Target Audience

Capacity Connect empowers all stakeholders in the workforce ecosystem with personalized learning, verified credentials and data-driven insights.



Trainees / Students

- ✓ Personalized learning paths
- ✓ Skill-gap analysis & recommendations
- ✓ Verified digital certificates
- ✓ Progress tracking & career growth



Trainers / Mentors

- ✓ Automated course management
- ✓ Rating & performance dashboard
- ✓ Easy student tracking
- ✓ Wider reach & more opportunities



Institutions / Enterprises

- ✓ Workforce analytics & insights
- ✓ Better resource planning
- ✓ Standardized skill development
- ✓ Improved placement outcomes

2 Key Benefits & Expected Outcomes



60%

COST REDUCTION

Reduces manual efforts and administrative expenditure.



80%

ADMIN TIME SAVED

Automates processes from registration to certification.



CERTIFICATE AUTHENTICITY

QR-based verification prevents fraud and builds trust.



3X

FASTER TRAINER ALLOCATION

AI-powered matching connects trainers with right learners.



100%

PAPERLESS ECOSYSTEM

Digital workflows eliminate paper-based processes and reduce administrative waste.



40%

FASTER SKILL GAP IDENTIFICATION

AI analytics identifies workforce deficiencies early.

3 Website Link & Github repository

 **GitHub** <https://github.com/teamprerak/capacity-connect>



<https://capacity-connect-web-seven.vercel.app/>



1

Global Workforce & Industry Research

(Problem Validation)



Key Findings

- Over 50% of global employees need reskilling by 2025–2027.
- Rapid technological disruption is reshaping job roles.

Reference

- World Economic Forum (Future of Jobs Report)

Gartner Talent & HR Research

- 70% of employers report a lack of critical skills for future roles.

<https://www.weforum.org/publications/the-future-of-jobs-report-2023/>

2

National Standards & Educational Frameworks



NSQF & NEP 2020

- Sets skill and competency levels
- Supports flexible learning
- Encourages multidisciplinary education

NDEAR (Digital Education)

- Standardized skill development
- Inclusive learning
- Future-ready workforce



These frameworks ensure standardized, inclusive and future-ready skill development in India.

3

Technical Architecture & Security Standards



OWASP Application Security Verification Standard (ASVS v4.0)

- Follows OWASP security standards
- Uses secure login with JWT
- Uses role-based access (RBAC)
- Adds multiple security layers

High-Concurrency Relational Modeling

- Planned architecture: PostgreSQL 15+ indexing and Prisma ORM.
- Designed to handle 40+ relational entities efficiently.



**Secure • Scalable • Future-Ready
Built with industry standards and best practices.**